

Data Processing and Analysis

Here are the factors to consider during data processing and analysis:

Calculating Performance Metrics:

Performance metrics are key indicators that provide a quantitative measure of a drone's efficiency and effectiveness. These metrics help assess how well the drone performs its intended functions under various conditions. Common performance metrics include:

- **Average Speed:** The mean speed of the drone during a flight, which is crucial for applications like delivery services where time efficiency is essential.
- **Maximum Acceleration:** The highest rate of change of velocity indicates the drone's responsiveness and agility.
- **Total Flight Time:** The duration the drone can stay airborne on a single battery charge, which is vital for understanding endurance and energy efficiency.
- **Path Accuracy:** The degree to which the drone follows its intended flight path is important for tasks like surveying and mapping where precision is required.

Example: Imagine you are testing a drone designed for a delivery service. After conducting a series of test flights, you gather the following data:

- **Flight 1:** Speed - 15 m/s, Flight Time - 20 minutes, Acceleration - 2 m/s²
- **Flight 2:** Speed - 14 m/s, Flight Time - 22 minutes, Acceleration - 1.8 m/s²
- **Flight 3:** Speed - 16 m/s, Flight Time - 18 minutes, Acceleration - 2.2 m/s²

To calculate the average speed, you sum the speeds of all flights and divide by the number of flights:

$$\text{Average Speed} = \frac{15+14+16}{3} = 15 \text{ m/s}$$



If the required specification for the delivery service is an average speed of 17 m/s, further analysis is needed. You may examine if battery limitations or motor performance are causing the lower-than-expected speeds.

Identifying Patterns, Trends, and Outliers:

By analysing data over multiple flights, you can identify consistent patterns and trends that reveal how different variables affect drone performance. Outliers, or data points that deviate significantly from others, can indicate unusual or unexpected behaviour that needs investigation.

Example: Consider you have collected flight time data under various weather conditions:

- **Calm Weather:** Flight 1 – 25 minutes, Flight 2 – 24 minutes, Flight 3 – 26 minutes
- **Windy Weather:** Flight 1 – 18 minutes, Flight 2 – 19 minutes, Flight 3 – 17 minutes

By comparing the average flight times, you notice a pattern:

- **Average Flight Time (Calm Weather):** $\frac{25+24+26}{3} = 25$ minutes
- **Average Flight Time (Windy Weather):** $\frac{18+19+17}{3} = 18$ minutes

This pattern shows that flight time significantly decreases in windy conditions. This trend suggests the drone consumes more power to maintain stability and counteract wind, indicating the need for more powerful motors or better aerodynamic design to improve performance in such conditions.

Outliers might appear as flights where the drone performs unexpectedly well or poorly. For instance, a single flight time of 30 minutes in calm weather would be an outlier compared to the other data points, prompting further investigation into what caused this anomaly.

Optimising Performance:

Using the insights gained from data analysis, you can make informed adjustments to improve the drone's design, software, or operational

procedures. Optimization aims to enhance performance, efficiency, and reliability based on empirical data.

Example: Suppose your analysis reveals that the drone's battery life is consistently shorter than expected, especially during windy conditions. To address this, consider the following optimization strategies:

1. **Higher-Capacity Batteries:** Switching to batteries with greater capacity can extend flight time. For example, replacing a 4000 mAh battery with a 6000 mAh battery might provide the necessary endurance.
2. **Optimising Flight Paths:** Adjusting flight paths to minimise power consumption. For instance, programming the drone to fly at optimal altitudes where wind resistance is lower can conserve battery life..
3. **Upgrading to More Efficient Motors:** Using motors that offer better efficiency can reduce power draw. For example, brushless motors are typically more efficient than brushed motors, providing longer flight times for the same battery capacity.
4. **Improving Aerodynamics:** Modifying the drone's design to reduce drag and improve stability in windy conditions. For instance, streamlining the airframe and adding stabilising fins can enhance aerodynamic performance..

By implementing these changes, you might achieve the following improvements:

- **Battery Upgrade:** Extends flight time from 18 minutes to 25 minutes in windy conditions.
- **Optimised Flight Paths:** Reduces average power consumption, increasing overall efficiency.
- **Efficient Motors:** Provides a 10% increase in flight time due to lower energy usage.
- **Aerodynamic Improvements:** Enhances stability, reducing power consumption needed for stabilisation.